THE SALVOES AND THE THWARTING

How various agencies countered, and plan to combat, different kinds of attacks

THE ATTACKS

Website defacements

Getting unauthorised access to websites and altering content, often with political/ nationalistic messages

Data breaches

Attempts to steal sensitive information from government agencies, businesses and individuals

Denial of Service (DoS) & Distributed Denial of Service (DDoS)

These attacks overwhelm a target server/ network with traffic, putting it out of use

Malware distribution

The use of viruses to infect systems and gain control, usually for data theft

Phishing

The use of deceptive emails/ messages to trick individuals into revealing sensitive information

APT

Advanced Persistent Threats (APTs) establish an undetected presence in targeted networks

THE DEFENCE

Hardening critical infrastructure

India ramped up cyber audits and real-time monitoring of power grids, defence networks, telecom etc. Sector-specific computer emergency response teams (CERTs) were activated

Offensive cyber-readiness

This involves pre-emptive and retaliatory cyber operations targeting known Pakistani hacker groups and command servers. Marks a shift from passive defence to active deterrence

Threat-Intelligence integration

CERT-In, NTRO and military intelligence created a joint cyber threat fusion cell to track and neutralise threats. Information-sharing with allies was scaled up

Cyber hygiene campaigns

Recognising Pakistan's focus on phishing, India launched targeted awareness drives across government offices, military and public sector units. Regular drills and simulated attacks are routine

Legal/ Diplomatic pushback

India is invoking international cyber norms to expose Pakistan's digital aggression in global forums. New legislation is in the works to empower agencies to prosecute foreign cyber attackers and freeze assets

for phishing targeting its organisations. Websites of Pakistan's Sindh Police and its airport systems were breached too.

However, most of India's robust response was defensive in nature. It involved tripling cyber defence teams, deploying real-time intelligence-sharing and activating a joint task force led by the Data Security Council of India, coordinating government, private firms and industry bodies. Measures included temporarily blocking vulnerable financial sites, issuing CERT-In alerts and monitoring suspicious command servers. According to Bhattacharya, the Indian government activated a multi-pronged cyber defence operation. The Indian Cyber Crime Coordination Centre (I4C), under the Union ministry of home affairs, spearheaded counter-hack operations, reportedly taking down over 150 hostile command-and-control servers. India's proactive approach and coordination at the macro level helped it withstand the dynamic threat landscape in this virtual war.



"These virtual invasions are no longer just for ransom.... They target critical infrastructures. Mass disturbances in anonymity make cyber warfare a preferred tool for hostile actors"

TARUN WIG
CEO, Innefu Labs

With cyber threats ever present and evolving, India cannot let its guard down. To effectively counter Chinese cyber attacks, Pakistani hacktivists and other hacking groups, India must expand AI-powered threat detection and real-time incident response systems and improve cyber-security training. Increased investment in quantum-resistant encryption, cloud security and resilient infrastructure for all vital sectors are critical.

Wig says India has responded to growing cyber threats by fortifying its cyber defence through CERT-In and NCIIPC, indigenous cyber-security solutions and collaboration between government and private sectors. The Digital Personal Data Protection Act, 2023 has strengthened regulatory frameworks, while partnerships with Singapore, Japan and the UK have enhanced threat intelligence-sharing. India's determination to protect its digital infrastructure will shape the future of cyber-security in the region. ■